the orbits of photons temporarily trapped in it, see Edward Teo's technical paper (Teo 2003).

Chapter 7. Gravitational Slingshots

For a discussion of gravitational slingshots at a modestly more technical level than mine, I recommend the Wikipedia article http://en.wikipedia.org/wiki/Gravity_assist. But don't believe what it says about slingshots around black holes. Its statement (as of July 4, 2014) that "if a spacecraft gets close to the Schwarzschild radius [horizon] of a black hole, space becomes so curved that slingshot orbits require more energy to escape than the energy that could be added by the black hole's motion" is just plain wrong. Indeed, you should always read Wikipedia with some cautious skepticism. In my experience, in areas where I am an expert, roughly 10 percent of Wikipedia's statements are wrong or misleading.

More reliable than Wikipedia for gravitational slingshots, but less comprehensive, is http://www2.jpl.nasa.gov/basics/grav/primer.php. A gravitational-slingshot video game has been developed in connection with *Interstellar*; see Game.InterstellarMovie.com.

For a somewhat technical discussion of the intermediate-mass black holes that I invoke for gravitational slingshots, see Chapter 4 of *Black Hole Astrophysics: The Engine Paradigm* (Meier 2012).

You can generate and explore complicated orbits around fast-spinning black holes, such as that in Figure 7.6, using a tool written by David Saroff and available at http://demonstrations.wolfram.com/3DKerrBlackHoleOrbits.

Chapter 8. Imaging Gargantua